

Solutions

1. matter → comprised of atoms (indivisible particles)
 - atoms differ in size, shape...
 - empty space between atoms.
 - Atoms in constant motion
 -
2. Earth → solids
Wind → gas
Fire → plasma
Water → liquids
3.
 - All matter is composed of tiny, indivisible particles called atoms
 - All atoms of an element have identical properties
 - Atoms of different elements have diff. Properties.
 - Atoms of two or more elements can combine in constant ratio to form new substances.
 - In chemical reactions, atoms join together or separate from each other but are not destroyed.
4. use overheads
5. $p^+ = 1 \text{ amu}$, $n^0 = 1 \text{ amu}$, $e^- = \text{negligible}$

Name	Symbol	Charge	Mass	Location
Proton	P	+1	1.7×10^{-24}	Nucleus
Neutron	N	0	1.7×10^{-24}	Nucleus
Electron	E	-1	Negligible (0)	Shells/orbitals

- 1 atomic mass unit (amu) = 1 U = 4.7×10^{-24} grams
 - the mass of one electron is 9.1×10^{-28} grams which is so small that we say it is negligible (in other words = zero)
6. use overhead
 7. same overhead as in 6
 8. nucleus contains neutrons
 9. Z #
 10. $\# p^+ = \# e^-$
 11. protons + neutrons
 12. overhead
 13. hydrogen, deuterium, tritium
 14. the percent of an isotope in a sample of an element.
 15. an unstable isotope that upon decay releases radiation.
 16. Brooks discovered that an atom can spontaneously change into another element.
 17. the time it takes for one-half the nuclei in a radioactive sample to decay
 18. technique that uses radioactive C-14 to identify the date of death of once-living material.
- Carbon-14 reacts with oxygen in the air to form CO_2 this is absorbed by plants and eventually makes its way into the food chain. When an organism dies there is no more consumption of this C-14. The C-14 in the dead organism starts to decay so by taking the ratio of carbon-14 in the dead organism to the amount one would expect to find in a living organism it can be dated.